

The randomization occurs in two places. First, the training dataset is randomized: each tree is built considering only a subset of the training set, which is randomly selected and will be different for every tree. (The test set is randomly selected at the beginning of the process, but it's not rerandomized or reselected for every tree.) Second, the variables used to build the tree are randomized: only a subset of the full set of variables is used for each tree, and the subset could be different every time as well.

After building a collection of these different, randomized trees, we have a whole random forest. To make a prediction about a particular observation, we have to find what each of these different decision trees predicts, and then take the average of the prediction for every individual decision tree. Since the decision trees are randomized in both their data and their variables, taking an average of all of them helps avoid the problem of overfitting and often leads to more accurate predictions.

Our code in this chapter creates decision trees “from scratch,” by directly manipulating datasets and lists and loops. When you work with decision trees and random forests in the future, you can rely on existing Python modules that do much of that heavy lifting for you. But don't let these modules become a crutch: if you understand every step of these important algorithms well enough to code them from scratch yourself, you can be much more effective in your machine learning efforts.

Summary

This chapter introduced machine learning and explored decision tree learning, a fundamental, simple, and useful machine learning method. Decision trees constitute a type of algorithm, and the generation of a decision tree is itself an algorithm, so this chapter contained an algorithm for generating an algorithm. By learning decision trees and the fundamental ideas of random forests, you have taken a big step toward becoming a machine learning expert. The knowledge you've gained in this chapter will be a solid foundation for other machine learning algorithms you may choose to learn, including advanced ones like neural networks. All machine learning methods attempt the type of task we tried here: prediction based on patterns in a dataset. In the next chapter, we explore artificial intelligence, one of the most advanced undertakings of our adventure.